

DSA0502 – QUERY PROCESSING FOR DATA SCIENCE

CO2 – AT1 – DESIGN AND CONNECTIVITY

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Experiment 1: Hotel Reservation and Booking System

Question

Develop a **Hotel Reservation and Booking System** using **Python and SQLite**. The system should maintain customer information, room details, reservations, and payments. It should demonstrate customer management, room booking, reservation updates, and invoice generation while maintaining a normalized database structure.

Aim:

To design and implement a **Hotel Reservation and Booking System** using **Python and SQLite** for managing customers, rooms, bookings, payments, reservation updates, and invoice generation.

Table 1: Customer

Field Name	Data Type	Description
ID	INTEGER (Primary Key)	Customer ID
Name	TEXT	Customer Name

Table 2: Room

Field Name	Data Type	Description
RoomNo	INTEGER (Primary Key)	Room Number
Type	TEXT	Room Category
Price	INTEGER	Room Price
Status	TEXT	Room Availability

Table 3: Booking

Field Name	Data Type	Description
ID	INTEGER (Primary Key)	Booking ID
CustomerID	INTEGER (Foreign Key)	Customer ID
RoomNo	INTEGER (Foreign Key)	Room Number

Field Name	Data Type	Description
CheckIn	TEXT	Check-in Date
CheckOut	TEXT	Check-out Date

Table 4: Payment

Field Name	Data Type	Description
ID	INTEGER (Primary Key)	Payment ID
BookingID	INTEGER (Foreign Key)	Booking ID
Amount	INTEGER	Payment Amount

Relationship Between Tables

Parent Table	Relationship	Child Table
Customer	1 : Many	Booking
Room	1 : Many	Booking
Booking	1 : 1	Payment

Program:

```
import sqlite3
```

```
# Connect Database
```

```
conn = sqlite3.connect("hotel.db")
```

```
cur = conn.cursor()
```

```
# Create Tables
```

```
cur.execute("CREATE TABLE IF NOT EXISTS Customer(ID INTEGER, Name TEXT)")
```

```
cur.execute("CREATE TABLE IF NOT EXISTS Room(RoomNo INTEGER, Type TEXT, Price INTEGER, Status TEXT)")
```

```
cur.execute("CREATE TABLE IF NOT EXISTS Booking(ID INTEGER, CustomerID INTEGER, RoomNo INTEGER, CheckIn TEXT, CheckOut TEXT)")
```

```
cur.execute("CREATE TABLE IF NOT EXISTS Payment(ID INTEGER, BookingID INTEGER, Amount  
INTEGER)")
```

```
# Clear Old Data
```

```
cur.execute("DELETE FROM Customer")
```

```
cur.execute("DELETE FROM Room")
```

```
cur.execute("DELETE FROM Booking")
```

```
cur.execute("DELETE FROM Payment")
```

```
# Insert Data
```

```
cur.execute("INSERT INTO Customer VALUES(1,'Rahul')")
```

```
cur.executemany("INSERT INTO Room VALUES(?,?,?)",[  
(101,"Standard",2000,"Available"),  
(102,"Deluxe",3000,"Available"),  
(103,"Suite",5000,"Available")  
])
```

```
# Book Room
```

```
cur.execute("INSERT INTO Booking VALUES(1,1,101,'2026-08-01','2026-08-03')")
```

```
cur.execute("UPDATE Room SET Status='Booked' WHERE RoomNo=101")
```

```
cur.execute("INSERT INTO Payment VALUES(1,1,6000)")
```

```
conn.commit()
```

```
# Show Booking
```

```
print("Booking Details")
```

```
cur.execute("""  
SELECT Name, RoomNo, CheckIn, CheckOut  
FROM Customer, Booking  
WHERE Customer.ID = Booking.CustomerID  
""")
```

```
for row in cur.fetchall():
```

```

        print(row)

# Update Booking
cur.execute("UPDATE Booking SET CheckOut='2026-08-04' WHERE ID=1")
conn.commit()

print("\nUpdated Booking")
cur.execute("SELECT * FROM Booking")
print(cur.fetchall())

# Show Invoice
print("\nInvoice")
cur.execute("""
SELECT Customer.Name, Payment.Amount
FROM Customer, Booking, Payment
WHERE Customer.ID=Booking.CustomerID
AND Booking.ID=Payment.BookingID
""")
for row in cur.fetchall():
    print(row)

conn.close()

```

OUTPUT:

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/balaj/OneDrive/Documents/hotel.py =====
Booking Details
('Rahul', 101, '2026-08-01', '2026-08-03')

Updated Booking
[(1, 1, 101, '2026-08-01', '2026-08-04')]

Invoice
('Rahul', 6000)
>>>
```

Experiment 2: Employee Payroll Management System

Question

Develop an Employee Payroll Management System using Python and SQLite. The system should maintain employee details, department information, attendance records, salary details, and payroll information. It should demonstrate employee management, payroll generation, salary updates, and payroll display using database connectivity.

Aim

To design and implement an Employee Payroll Management System using Python and MySql for managing employee details, departments, attendance, salary, and payroll generation.

Database Tables

Table 1: Employee

Field Name	Data Type	Description
ID	INT (Primary Key)	Employee ID
Name	VARCHAR(50)	Employee Name
Salary	INT	Employee Salary

Table 2: Payroll

Field Name	Data Type	Description
ID	INT (Primary Key, Foreign Key)	Employee ID
Tax	INT	Tax Deduction
NetSalary	INT	Salary after Tax

Relationship

Entity 1	Relationship	Entity 2
Employee	1 : 1	Payroll

Program

```
import mysql.connector
```

```
# Connect to MySQL
```

```
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="kirth123",
    database="employee_payroll"
)
```

```
cur = conn.cursor()
```

```
# Create Tables
```

```
cur.execute("""
CREATE TABLE IF NOT EXISTS Employee(
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Salary INT
)
""")
```

```

cur.execute("""
CREATE TABLE IF NOT EXISTS Payroll(
    ID INT PRIMARY KEY,
    Tax INT,
    NetSalary INT
)
""")

# Delete Old Data
cur.execute("DELETE FROM Employee")
cur.execute("DELETE FROM Payroll")

# Insert Employee
cur.execute("INSERT INTO Employee VALUES(1,'Rahul',30000)")

# Calculate Payroll
salary = 30000
tax = 3000
net = salary - tax

cur.execute("INSERT INTO Payroll VALUES(%s,%s,%s)", (1, tax, net))

conn.commit()

# Display Employee Details
print("Employee Details")
cur.execute("SELECT * FROM Employee")
for row in cur.fetchall():
    print(row)

```

```

# Update Salary

cur.execute("UPDATE Employee SET Salary=32000 WHERE ID=1")

cur.execute("UPDATE Payroll SET Tax=3200, NetSalary=28800 WHERE ID=1")

conn.commit()


# Display Payroll

print("\nPayroll Details")

cur.execute("""

SELECT Employee.Name, Employee.Salary, Payroll.Tax, Payroll.NetSalary

FROM Employee

JOIN Payroll

ON Employee.ID = Payroll.ID

""")


for row in cur.fetchall():

    print(row)


conn.close()

```

OUTPUT:

```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/balaj/OneDrive/Documents/sql_connect.py =====
Employee Details
(1, 'Rahul', 30000)

Payroll Details
('Rahul', 32000, 3200, 28800)
>>> |

```